2.3 Pre-Quiz: Sequences, Lines, Radicals and Exponents — Markscheme

1. [Maximum mark: 6]

(a) $r = \frac{12}{24} = \frac{1}{2}$ **A1**

(b) $u_6 = 24\left(\frac{1}{2}\right)^5$ **M1**

$u_6 = \frac{24}{32} = \frac{3}{4}$ **A1**

Accept 0.75. *FT from (a).*

(c) $|r| < 1 \quad$ (since $r = \frac{1}{2}$) **R1**

Accept "$-1 < r < 1$*" or "r is between* -1 *and* 1*". Do not accept "the terms get smaller" alone.*

(d) $\frac{10}{1-r} = 15$ **M1**

$1 - r = \frac{10}{15} = \frac{2}{3}$

$r = \frac{1}{3}$ **A1**

2. [Maximum mark: 9]

(a) $M = \left(\frac{-4+6}{2},\ \frac{1+(-3)}{2}\right)$

$M = (1,\ -1)$ **A1A1**

A1 for each coordinate.

(b) $m_{PQ} = \frac{-3-1}{6-(-4)}$ **M1**

$m_{PQ} = -\frac{4}{10} = -\frac{2}{5}$ **A1**

(c) $m_L = \frac{5}{2}$ **A1**

FT from (b): the negative reciprocal of their gradient.

(d) $y - (-1) = \frac{5}{2}(x - 1)$ **M1**

$y + 1 = \frac{5}{2}(x - 1)$ **A1**

M1 for substituting their gradient and their M into any form of the equation of a line. Accept $y = \frac{5}{2}x - \frac{7}{2}$ or $5x - 2y = 7$. FT from (a) and (c).

(e) $9 + 1 = \frac{5}{2}(k - 1)$ **M1**

$k - 1 = 4$

$k = 5$ **A1**

M1 for substituting $(k, 9)$ into their equation of L. FT from (d).

3. [Maximum mark: 12]

(a) y-intercept -3 **A1**

$2x - 3 = 0$, so $x = \frac{3}{2}$ **A1**

Accept $(0, -3)$ *and* $\left(\frac{3}{2}, 0\right)$.

(b) gradient of g is 2 **A1**

$g(x) = 2x$ **A1**

The gradient may be seen in the expression.

(c) gradient of h is $-\frac{1}{2}$ **A1**

$y - 6 = -\frac{1}{2}(x - 2)$ **M1**

$h(x) = -\frac{1}{2}x + 7$ **A1**

M1 for substituting their gradient and $(2, 6)$ *into any form of the equation of a line. Accept* $h(x) = -\frac{1}{2}(x - 2) + 6$.

(d) $2x - 3 = -\frac{1}{2}x + 7$ **M1**

$\frac{5}{2}x = 10$

$x = 4$ **A1**

$y = 2(4) - 3 = 5$, so the point is $(4,\ 5)$ **A1**

FT from (c).

(e) $-\frac{1}{2}x + 7 = 0$ **M1**

$R = (14,\ 0)$ **A1**

Accept $x = 14$. *FT from (c).*

4. [Maximum mark: 13]

(a) $4^x = (2^2)^x = 2^{2x}$ **M1**

$2^{2x} \times 2^{x+1} = 2^{3x+1}$

$a = 3, \quad b = 1$ **A1A1**

(b) $2^{3x+1} = 2^4$ **M1**

$3x + 1 = 4$

$x = 1$ **A1**

M1 for writing 16 *as* 2^4 *and equating exponents. FT from (a).*

(c) $(3^2)^x = 3^3$ **M1**

$2x = 3$ **A1**

$x = \frac{3}{2}$ **A1**

M1 for writing both sides as powers of 3*. Accept* $x = 1.5$*.*

(d) $\sqrt{12} = 2\sqrt{3}$ **M1**

$\left(2\sqrt{3} + \sqrt{3}\right)^2 = \left(3\sqrt{3}\right)^2$

$= 9 \times 3$ **A1**

$= 27$ **AG**

Accept expansion: $12 + 2\sqrt{36} + 3$ *(M1),* $12 + 12 + 3$ *(A1).*

(e) $\frac{6}{3-\sqrt{3}} \times \frac{3+\sqrt{3}}{3+\sqrt{3}}$ **M1**

$= \frac{6\left(3+\sqrt{3}\right)}{9-3}$ **A1**

$= 3 + \sqrt{3}$ **A1**

A1 for the denominator 6*. Accept* $a = 3$*,* $b = 1$*.*